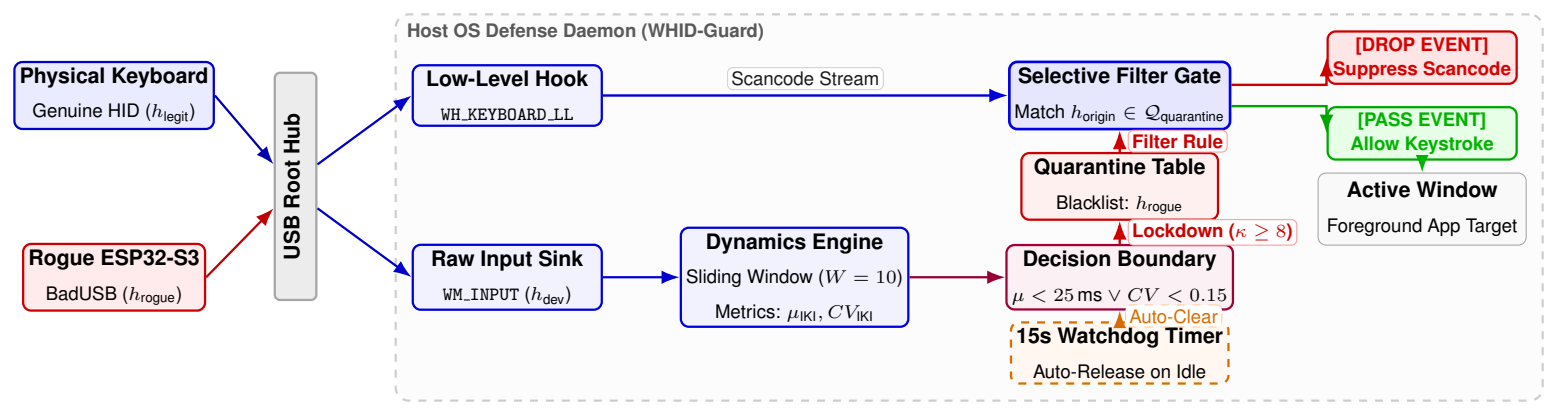

Host OS Defense Daemon (WHID-Guard)
Physical Keyboard
Genuine HID (h_{legit})
Rogue ESP32-S3
BadUSB (h_{rogue})
USB Root Hub
Low-Level Hook
WH_KEYBOARD_LL
Raw Input Sink
WM_INPUT (h_{dev})
Scancode Stream
Dynamics Engine
Sliding Window ($W = 10$)
Metrics: $\mu_{\text{IKI}}, CV_{\text{IKI}}$
Selective Filter Gate
Match $h_{\text{origin}} \in \mathcal{Q}_{\text{quarantine}}$
Filter Rule
Quarantine Table
Blacklist: h_{rogue}
Lockdown ($\kappa \geq 8$)
Decision Boundary
$\mu < 25\,\text{ms} \vee CV < 0.15$
Auto-Clear
15s Watchdog Timer
Auto-Release on Idle
[DROP EVENT]
Suppress Scancode
[PASS EVENT]
Allow Keystroke
Active Window
Foreground App Target